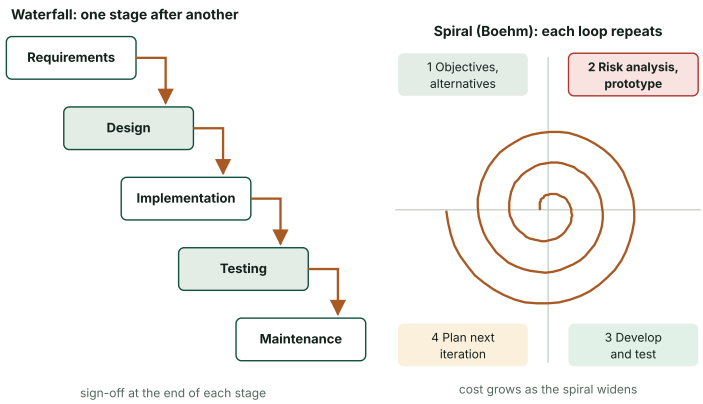
The waterfall model, iterative and spiral development

AP10 FACT FILE

Specification: students should be able to

- Describe the main features of the waterfall model of systems development
- Describe each stage of the waterfall model and the deliverable it produces
- Describe the involvement of users when a system is developed using the waterfall model
- Describe iterative development and the spiral model, including the role of risk analysis
- Evaluate the waterfall model, including its advantages and disadvantages, and its suitability for different projects

Key words: development methodology, waterfall model, sequential, deliverable, sign-off, system specification, design document, handover, maintenance, iterative development, iteration, increment, spiral model, risk analysis, prototype, stable requirements, change control



The waterfall model as a cascade of sequential stages beside the spiral model with its four quadrants and risk analysis in every loop

Overview: why development models matter

Turning an identified need into a working, installed system is a complex process, and organisations follow a development methodology to give it structure. There is no single perfect methodology. Each one makes different choices about how strictly the stages are ordered, how often users are involved and how easily requirements can change, so each suits some projects better than others. The waterfall model is the traditional approach: a fixed sequence of stages in which each must be completed and signed off before the next begins, producing a deliverable that feeds the following stage. Iterative approaches instead build the system in repeated cycles, so that users can see and comment on working versions early. The spiral model is an iterative approach that places risk analysis at the centre of every cycle. Understanding the features, strengths and weaknesses of each model allows you to recommend a suitable approach for a particular project, which is exactly what examination questions in this area ask you to do.

The stages of the waterfall model

The waterfall model is usually shown as five stages. In the requirements or analysis stage the systems analyst uses fact-finding methods (interviews, observation, questionnaires and inspecting documents) to understand any existing system and the needs of its users. The deliverable is the system specification, which must be clear, unambiguous and complete, because a requirement missed here is difficult and expensive to add later. In the design stage the specification is turned into a design document describing the processes, their inputs and outputs, the software units and how they interact, the user interface and the hardware needed. It must be precise because programmers code directly from it. In implementation a team of programmers, each given specific tasks, writes the code to the design, unit testing as they go. In the testing stage the complete system is verified against the specification using a test plan. Finally, maintenance covers installation, migrating data from any old system, ongoing support and corrections or improvements over the life of the system.

Advantages and disadvantages of the waterfall model

The waterfall model divides work into distinct stages with clearly defined deliverables. This makes progress easy to measure, allows each stage to be signed off formally, lets the developer invoice the client stage by stage and produces thorough documentation that helps later maintenance. Because each stage needs different skills, specialist staff can move from one project to the same stage of another project. When requirements are stable and well understood, costs and timescales can be estimated with some confidence. The disadvantages are serious, however. The model is inflexible: once a stage is signed off, changing it means going back up the waterfall. Each handover between stages is a point at which misunderstandings can be introduced, and such errors may travel through several stages before anyone notices, often not until testing, when correcting them is most expensive. Users see no working software until near the end. Staff who specialise in a single stage are also less flexible than the multi-skilled teams used in Agile development, so workloads are harder to balance across projects.

Users and the waterfall model

In a waterfall project the users are mainly involved at two points. At the start, the analyst gathers their requirements, which is especially important when there is no existing system to investigate, because the users' own descriptions are then the main source of information. Near the end, users take part in acceptance testing to confirm that the finished system works as they intended. Between these points they usually see nothing working, so a misunderstanding about what they wanted can remain hidden for months. Many well-documented real project failures share the same pattern: the people who would actually use the system were not consulted properly, deliverables were not formally signed off, changes were not tracked, quality assurance was skipped and no test plans were written. The lesson is that the waterfall model only works if it is followed with discipline. Involving all stakeholders, formal sign-off, change control, quality assurance during coding and a test plan written from the specification all reduce the risk that a misunderstanding reaches the final system.

Iterative development

Iterative development builds a system through repeated cycles rather than a single pass. Each iteration includes some planning, analysis, design, coding and testing, and produces a working version or increment that users can try. Their feedback is used to refine the requirements and to decide what the next iteration should include. Because users see real software early, misunderstandings come to light while they are still cheap to correct, and useful parts of the system can be delivered before the whole project is complete. Iterative development therefore suits projects where requirements are not fully known at the start or are likely to change. It does have costs: it depends on users being available to give regular feedback, it can be harder to agree a fixed price and deadline in advance, and the scope can keep growing if every iteration adds new requests without control. Agile methods such as Scrum, covered in the next lesson, are modern iterative approaches with defined roles and events that help to manage these problems.

The spiral model

The spiral model, proposed by Barry Boehm in the 1980s, is an iterative model driven by risk. The project is drawn as a spiral that starts small at the centre and widens as time and cost increase. Each loop passes through four quadrants. First, the objectives, alternative approaches and constraints for that loop are determined. Second, the alternatives are evaluated and the risks are identified and resolved, often by building a prototype to test an uncertain idea. Third, the next level of the product is developed and tested. Fourth, the client reviews progress and the next loop is planned. Because the biggest risks are tackled first, before large amounts of money are committed, the spiral model suits large, complex, high-risk or novel projects, such as safety-related control systems. Its weaknesses are that risk analysis requires specialist expertise and adds cost, the project can be hard to manage and to bring to a close, and the overhead is not justified for small, low-risk developments.

T Key terms from the mark schemes

Waterfall model
A sequential development methodology in which distinct stages are completed in order, each producing a deliverable and being signed off before the next stage begins.
`Credited: Linear sequential model`

Deliverable
The end product of a stage of development, such as a system specification or design document, which is passed on as the input to the next stage.
`Credited: End product`

System specification
The deliverable of the requirements stage: a clear, unambiguous and complete document of the user requirements for the new system.
`Credited: Requirements specification`

Handover
The point at which work passes from one waterfall stage, and often one group of staff, to the next; misunderstandings introduced here may travel undetected through later stages.
`Credited: Handover interface`

Spiral model
A risk-driven iterative model in which each loop determines objectives, analyses and resolves risks, develops and tests the next version, and plans the next loop.
`Credited: Boehm's spiral model`

Sign-off
Formal agreement by the client or manager that a stage's deliverable is complete and accepted, allowing the next stage to begin.

Design document
The deliverable of the design stage, describing processes, inputs and outputs, software units, user interfaces and hardware precisely enough for programmers to code from.

Iterative development
An approach in which a system is built through repeated cycles of planning, design, coding and testing, each producing a version that users review.

Risk analysis
Identifying possible threats to a project's success, assessing their likelihood and impact, and deciding how to reduce or resolve them.

Exam focus

Questions on this lesson often ask you to outline the key characteristics of the waterfall model, briefly describe each stage, describe user involvement, or list advantages and disadvantages, typically for 1–2 marks per point. For features, mark schemes credit: a sequence of distinct stages, each completed before the next begins, a deliverable at the end of each stage, and the possibility of returning to an earlier stage if an error is found. When describing a stage, name its deliverable (system specification, design document) because that is often the creditworthy detail. For user involvement, say precisely when users are involved – requirements and acceptance testing – not just that involvement is 'low'. Disadvantage answers earn more when they explain consequences, for example that a misunderstanding at a handover may go undetected until testing, when it is costly to fix. For the spiral model, examiners look for iteration and risk analysis. In evaluation questions, finish with a recommendation tied to the scenario: stability of requirements, size, risk and user availability.

? Check yourself

1. State two stages of the waterfall model and the deliverable produced by each. [2]
2. A software firm in Ballymena is using the waterfall model to build a payroll system for a haulage company. Describe the key characteristics of the way this project will be run. [4]
3. Describe what happens during the design stage of the waterfall model. [3]

Answers: 1. Any two stage + deliverable pairs: requirements/analysis → system specification (1) / design → design document (1) / implementation → program code (1) / testing → test results / tested system (1) / maintenance → updated, supported system (1). Both stage and deliverable needed for each mark. 2. Any four from: consists of a sequence of distinct stages (1) / e.g. analysis, design, implementation, testing, maintenance (1) / each stage must be completed/signed off before the next begins (1) / each stage produces a deliverable (1) / the deliverable is the input to the next stage (1) / the team may have to return to an earlier stage if a later stage reveals a mistake (1) / user mainly involved at analysis and testing (1). 3. Any three from: starts from the system specification (1) / identifies the processes and the inputs and outputs of each (1) / designs the software modules and how they communicate with one another (1) / designs the user interface, e.g. storyboards (1) / identifies hardware needs (1) / produces a precise, unambiguous design document (1).